

Retinal AI Handbook

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Preface

The retina is the only part of the central nervous system that can be directly visualized non-invasively. Owing to its shared embryological origin, anatomical structure, and physiological function with the brain, retinal microvascular alterations have been associated with various systemic diseases. Advances in retinal imaging and artificial intelligence have transformed the retina from an organ primarily examined for ocular diseases into a valuable source of quantitative biomarkers for systemic health. Recent advances in artificial intelligence (AI) have further enhanced the potential of retinal imaging as a non-invasive biomarker for cerebrovascular health. These developments have transformed retinal image analysis from a purely qualitative clinical assessment to large-scale, quantitative, and automated evaluation of retinal images, enabling early detection of disease, risk stratification, and monitoring of disease progression.

The motivation for this handbook originated from my own experience as a Computer Science PhD student entering the field of retinal imaging. Although building machine learning models was familiar, understanding biomedical terminology, retinal anatomy, clinical concepts, and disease mechanisms proved to be a much greater challenge than expected. Many important concepts were scattered across ophthalmology textbooks, neuroscience references, epidemiology papers, and artificial intelligence literature. This fragmentation made it difficult to build a coherent understanding of the field. This handbook was therefore developed as a living knowledge base that helps translate complex clinical concepts into computational understanding. It is intended to evolve alongside scientific advances, serving as a living knowledge base for researchers, clinicians, and students exploring the intersection of retinal imaging, vascular biology, and artificial intelligence. The handbook is organized into chapters, each focusing on a specific topic, with a glossary of terms provided for reference. The content is written in English, with Vietnamese translations provided for selected glossary terms to facilitate understanding for Vietnamese-speaking readers. Rather than providing definitive answers, this handbook aims to support continuous learning, critical thinking, and interdisciplinary research throughout the rapidly evolving field of retinal imaging and artificial intelligence.

How to read this handbook

Purpose of This Handbook

The primary purpose of this handbook is to facilitate interdisciplinary learning by organizing retinal imaging knowledge into a structured knowledge base. Rather than providing definitive answers or isolated facts, this handbook emphasizes the relationships among biological concepts, clinical reasoning, biomedical terminology, and computational methods. By progressively connecting these concepts, the handbook aims to help readers develop a coherent understanding of retinal imaging research, allowing them to interpret scientific literature, communicate across disciplines, and conduct evidence-informed research.

Scope

This handbook focuses on the foundational concepts required to understand retinal imaging research from both biomedical and computational perspectives. This edition focuses primarily on color fundus photography (CFP), retinal microvasculature, retinal biomarkers, and their applications in systemic diseases, particularly stroke and atherosclerosis. To maintain conceptual consistency, this handbook prioritizes depth over breadth, focusing on concepts that form the foundation of retinal imaging research.

Future editions will progressively expand the handbook to include additional retinal imaging modalities, including optical coherence tomography (OCT), OCT angiography (OCTA), and fluorescein angiography (FA), together with multimodal retinal imaging, foundation models, and emerging computational approaches for retinal image analysis. Future editions will also explore the relationships between retinal imaging and systemic diseases, including diabetes, hypertension, and neurodegenerative disorders.

As retinal imaging research continues to evolve, the scope of this handbook will be updated accordingly to incorporate emerging scientific evidence and technological advances.